

Key

Comments in blue
Warnings in red

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Rules

- The cover page should only have your name and email. Nothing else may may be written on the cover page.
- No calculators.
- No notes, books, or other outside resources.
- No phones, smartwatches, earbuds, or any other device that communicates.
- Eyes on your own exam.
- Scratch paper is available upon request. You may only use the provided scratch paper, and all scratch paper (used and unused) must be submitted with your exam.

Expectations

- Write with proper notation.
- Show your work in order to earn points. University mathematics focuses on the **process** more than the **answer**. For example, stating “does not exist” without giving a reason will not earn points.
- *There is no guarantee that the practice exam has exactly all the same types of problems as the actual exam. (However, the practice exam is a good study resource for the actual exam.)*

1. Find $\lim_{x \rightarrow 3} \frac{8}{|x-3|}$

$$\lim_{x \rightarrow 3} 8 = 8$$

$$\lim_{x \rightarrow 3} |x-3| = |3-3| = 0$$

Do not write $\frac{8}{0}$ (frowny face)

x	y
2.9	$\frac{8}{ 2.9-3 } = \frac{8}{ -0.1 } = \frac{8}{0.1} = 80$
2.99	800

x	y
3.1	$\frac{8}{ 3.1-3 } = \frac{8}{ 0.1 } = \frac{8}{0.1} = 80$
3.01	800

$$\lim_{x \rightarrow 3^-} \frac{8}{|x-3|} = +\infty$$

$$\lim_{x \rightarrow 3^+} \frac{8}{|x-3|} = +\infty$$

plus sign optional

$$\lim_{x \rightarrow 3} \frac{8}{|x-3|} = +\infty$$

2. Find $\lim_{p \rightarrow 2} \frac{p^2 - 4}{p - 2}$

I did this first

$$\lim_{p \rightarrow 2} (p^2 - 4) = 2^2 - 4 = 0$$

$$\lim_{p \rightarrow 2} (p - 2) = 0$$

then this next

notice no limit notation here

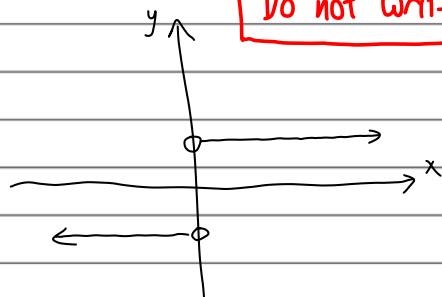
$$= \lim_{p \rightarrow 2} \frac{(p+2)(p-2)}{p-2} = \lim_{p \rightarrow 2} (p+2) = 2+2 = 4$$

Do not write $\frac{0}{0}$

Of course, you can skip from this to this.

3. Find $\lim_{x \rightarrow 0} \frac{x}{|x|}$

Do not write $\frac{0}{0}$



$$\lim_{x \rightarrow 0^-} \frac{x}{|x|} = -1$$

$$\lim_{x \rightarrow 0^+} \frac{x}{|x|} = 1$$

limits obtained from the graph

$$\lim_{x \rightarrow 0^-} \frac{x}{|x|} \neq \lim_{x \rightarrow 0^+} \frac{x}{|x|} \text{ so } \lim_{x \rightarrow 0} \frac{x}{|x|} \text{ dne}$$

4. Find $\lim_{x \rightarrow 0} x^2 \sin(\frac{1}{x})$

Do not write $O^2 \sin(\frac{1}{x})$

$$-1 \leq \sin(\frac{1}{x}) \leq 1$$

$$\underbrace{-x^2}_{f(x)} \leq \underbrace{x^2 \sin(\frac{1}{x})}_{g(x)} \leq \underbrace{x^2}_{h(x)}$$

$$\lim_{x \rightarrow 0} \underbrace{(-x^2)}_{f(x)} = -(0)^2 = 0$$

$$\lim_{x \rightarrow 0} \underbrace{(x^2)}_{h(x)} = 0^2 = 0$$

By the Squeeze Theorem, $\lim_{x \rightarrow 0} x^2 \sin(\frac{1}{x}) = 0$
 Must state by name $g(x)$

My blue comments are to try to match pieces with the book's notation about Squeeze Theorem.

5. Find $\lim_{x \rightarrow -\infty} \frac{(4x^5 + 6)\frac{1}{x^8}}{(7x^8 + 9)\frac{1}{x^8}}$
 whether $\rightarrow +\infty$ or $\rightarrow -\infty$ times top/bottom by $\frac{1}{x^8}$ bottom power

$$= \lim_{x \rightarrow -\infty} \frac{\frac{4}{x^3} + \frac{6}{x^8}}{7 + \frac{9}{x^8}}$$

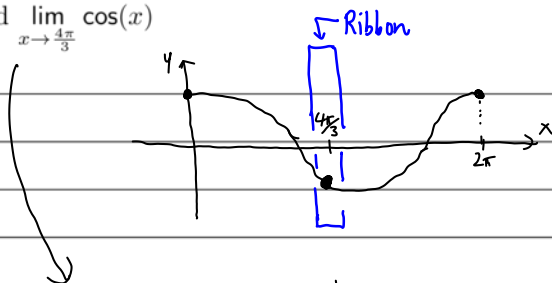
This work says we can "split"

$$= \frac{\lim_{x \rightarrow -\infty} (\frac{4}{x^3} + \frac{6}{x^8})}{\lim_{x \rightarrow -\infty} (7 + \frac{9}{x^8})}$$

Side $\lim_{x \rightarrow -\infty} (7 + \frac{9}{x^8}) = (\lim_{x \rightarrow -\infty} 7) + (\lim_{x \rightarrow -\infty} \frac{9}{x^8})$
 $= 7 + 9(\lim_{x \rightarrow -\infty} \frac{1}{x^8})$
 Okay to skip from here to here $\Rightarrow 7 + 9 \cdot 0$
 (but I showed more, to clarify) $= 7$

$$= \frac{4 \cdot 0 + 6 \cdot 0}{7 + 9 \cdot 0} = \frac{4}{7}$$

6. Find $\lim_{x \rightarrow \frac{4\pi}{3}} \cos(x)$



$$= \cos(\frac{4\pi}{3}) = -\frac{1}{2}$$

7. Find $\lim_{x \rightarrow 2} \frac{x+6}{x+3}$

$$= \frac{\lim_{x \rightarrow 2} (x+6)}{\lim_{x \rightarrow 2} (x+3)}$$

$$= \frac{8}{5}$$

I did this first. Then did this

$$\lim_{x \rightarrow 2} (x+6) = 2+6 = 8$$

Skipped (but I wanted to point out exactly when limit notation stops)

$$\lim_{x \rightarrow 2} (x+3) = 2+3 = 5$$

8. Find $\lim_{c \rightarrow 0} \frac{c^2}{(\sqrt{c^2+100}-10) \cdot (\sqrt{c^2+100}+10)}$

$$= \lim_{c \rightarrow 0} \frac{c^2(\sqrt{c^2+100}+10)}{c^2+100-100}$$

$$= \lim_{c \rightarrow 0} \frac{c^2(\sqrt{c^2+100}+10)}{c^2}$$

$$= \lim_{c \rightarrow 0} (\sqrt{c^2+100}+10)$$

$$= \sqrt{0^2+100}+10 = 10+10 = 20$$

Do not write $\frac{0}{0}$

9. Let $f(x) = x^2 + 2x + \sin(\pi x)$. Prove that there exists a number c in $(4, 10)$ such that $f(c) = 48.75$. Optional reading in footnote¹.

$f(x)$ is continuous on $[4, 10]$.

$$f(4) = 4^2 + 2(4) + \sin(4\pi) = 16 + 8 + 0 = 24$$

$$f(10) = (10)^2 + 2(10) + \sin(10\pi) = 100 + 20 + 0 = 120$$

Pick $N = 48.75$ which is between 24 and 120.

State by name
(or abbreviate IVT)

By the Intermediate Value Theorem, there is a number c in $(4, 10)$ such that $f(c) = 48.75$

¹If $f(x)$ represents the temperature of an object x hours after midnight, the work completed proves that some time between 4 hours after midnight and 10 hours after midnight, the object had a temperature of 48.75 degrees.

10. Given $f(x) = x^2 + 10x$ find $f'(3)$ directly by a limit-based definition. (No credit will be given for applying derivative rules such as the Power Rule.) *Optional reading in footnote².*

$$f'(3) = \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0} \frac{(3+h)^2 + 10(3+h) - (3^2 + 10 \cdot 3)}{h} = \lim_{h \rightarrow 0} \frac{9 + 6h + h^2 + 30 + 10h - 9 - 30}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2 + 16h}{h} = \lim_{h \rightarrow 0} \frac{h(h+16)}{h} = \lim_{h \rightarrow 0} (h+16) = 0+16 = 16$$

Do not write $\frac{0}{0}$

11. Find r and s so that

$$f(x) = \begin{cases} x^2 - 2r + 1 & \text{if } x < 5 \\ 3x + s & \text{if } x = 5 \\ 6r - 2x - 12 & \text{if } x > 5 \end{cases}$$

is continuous at $x = 5$, applying the definition.

Need $f(5)$ to exist $f(5) = 3(5) + s = 15 + s$ (For now, s can be any real.)

Need $\lim_{x \rightarrow 5^-} f(x)$ to exist $\lim_{x \rightarrow 5^-} f(x) = \lim_{x \rightarrow 5^-} (x^2 - 2r + 1) = 5^2 - 2r + 1 = 26 - 2r$

$$\lim_{x \rightarrow 5^+} f(x) = \lim_{x \rightarrow 5^+} (6r - 2x - 12) = 6r - 2(5) - 12 = 6r - 22$$

$$6r - 22 = 26 - 2r$$

$$8r = 48$$

$$r = 6$$

Need $\lim_{x \rightarrow 5} f(x) = f(5)$

$$14 = 15 + s$$

$$-1 = s$$

If $r = 6$, the limit from left and right are both 14, making $\lim_{x \rightarrow 5} f(x) = 14$

Conclusion: Must have $r = 6$ and $s = -1$

12. Find a horizontal asymptote of $f(x) = \frac{5x^2 + 8}{6x^2 - 1}$

We can do either $\lim_{x \rightarrow \infty} f(x)$ or $\lim_{x \rightarrow -\infty} f(x)$

$$\lim_{x \rightarrow \infty} \frac{5x^2 + 8}{6x^2 - 1} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x^2}(5x^2 + 8)}{\frac{1}{x^2}(6x^2 - 1)} = \lim_{x \rightarrow \infty} \frac{5 + \frac{8}{x^2}}{6 - \frac{1}{x^2}}$$

Can we "split"? Let's check.

$$\lim_{x \rightarrow \infty} (6 - \frac{1}{x^2}) = 6 - 1 \cdot 0 = 6$$

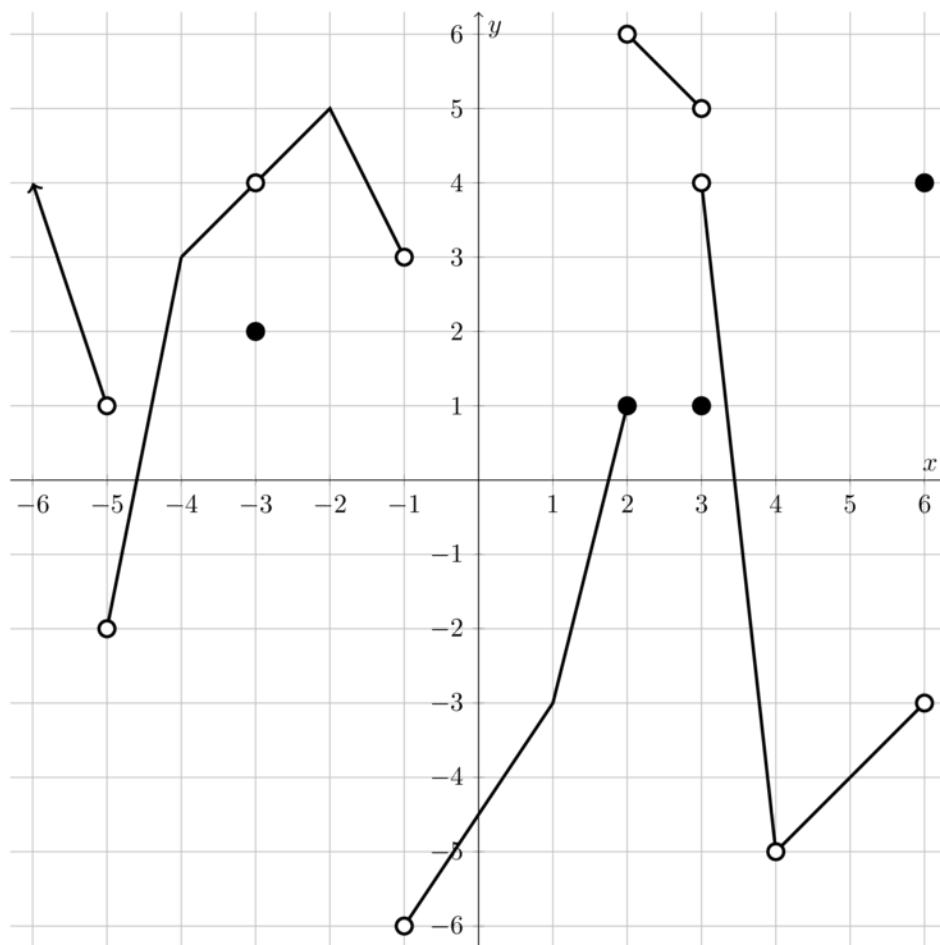
Not zero
so yes

$$= \frac{\lim_{x \rightarrow \infty} (5 + \frac{8}{x^2})}{\lim_{x \rightarrow \infty} (6 - \frac{1}{x^2})} = \frac{5 + 8 \cdot 0}{6 - 1 \cdot 0} = \frac{5}{6}$$

So $y = \frac{5}{6}$ is a horizontal asymptote

²If $f(x)$ represents the position (in meters) of a boat x minutes after midnight, then $f'(3)$ finds the velocity of the boat (in meters per minute) 3 minutes after midnight.

13. Based on the graph below for $y = f(x)$ answer all parts. As a reminder, if a requested value does not exist, provide a brief explanation:



(a) Find $f(-3) + 10 \cos(\pi)$
 $= 2 + 10(-1) = -8$

(b) Find $\lim_{x \rightarrow 6^-} f(x) = -3$

(c) Find $\lim_{x \rightarrow -3} f(x) = 4$

(d) Find $f(6) + \lim_{x \rightarrow 4} f(x) = 4 + (-5) = -1$

(e) Find $\lim_{x \rightarrow -5} f(x)$ dne because $\lim_{x \rightarrow -5^-} f(x) \neq \lim_{x \rightarrow -5^+} f(x)$

(f) Find $f(3) + \lim_{x \rightarrow 1} f(x) = 1 + (-3) = -2$

(g) Find $\tan\left(\frac{\pi}{4}\right) + \lim_{x \rightarrow 3^+} f(x) = 1 + 4 = 5$
 (Note: $\lim_{x \rightarrow 3^+}$ is from the right only)

(h) Find $f(6) + \lim_{x \rightarrow 1} \ln(x) = 4 + 0 = 4$

